

# Multimodal RAG

Chat with Text, Tables, Images & Graphs



TEXT



TABLES



IMAGES



GRAPHS

# A real PDF is not just text

Enterprise documents mix multiple modalities on the same page.

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THE PROBLEM



## TEXT

Policies, summaries, descriptions, notes



## TABLES

KPIs, regional metrics, financial data



## IMAGES

Products, facilities, screenshots, diagrams



## GRAPHS

Trends, comparisons, percentages, time series



## One report can contain all four at once

Example: NovaCore’s financial page contains narrative explanation, a structured financial table, and a quarterly revenue graph. A useful RAG system should reason over all of them—not only the paragraph text.

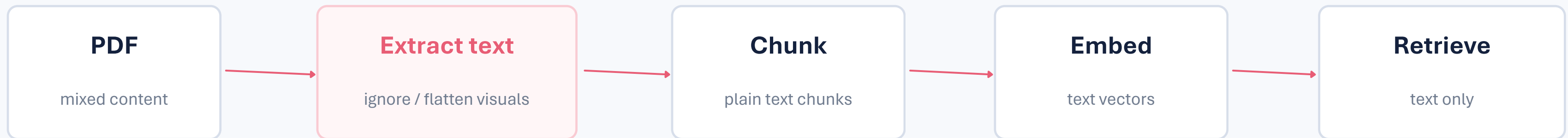
KEY IDEA

The user should be able to ask one question across any modality.

# Why traditional text RAG is not enough

The classic pipeline often loses structure and visual meaning.

## 2 TRADITIONAL RAG LIMITATION



×

### Example A — Chart

Question: “Which quarter had the highest revenue?”

Problem: The answer is encoded in the graph, not in ordinary paragraph text.

×

### Example B — Diagram

Question: “Where is the critical quality-control point?”

Problem: The relationship is inside a supply-chain diagram.

×

### Example C — Table

Question: “Which region has the highest CSAT?”

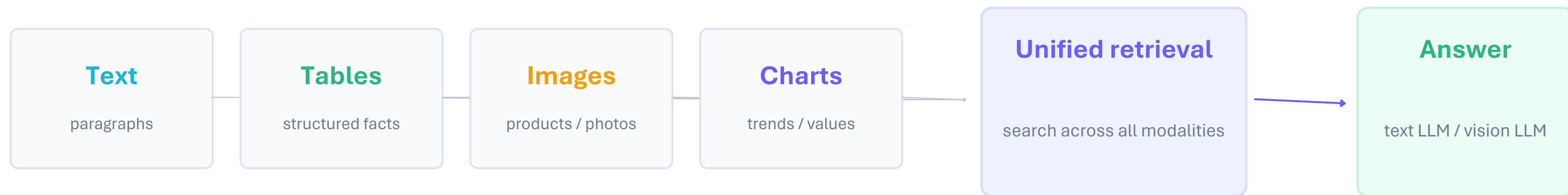
Problem: Flattened table text can lose row/column relationships.

# What is Multimodal RAG?

RAG that retrieves and reasons over more than one type of information.

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## THE SOLUTION



### Simple definition

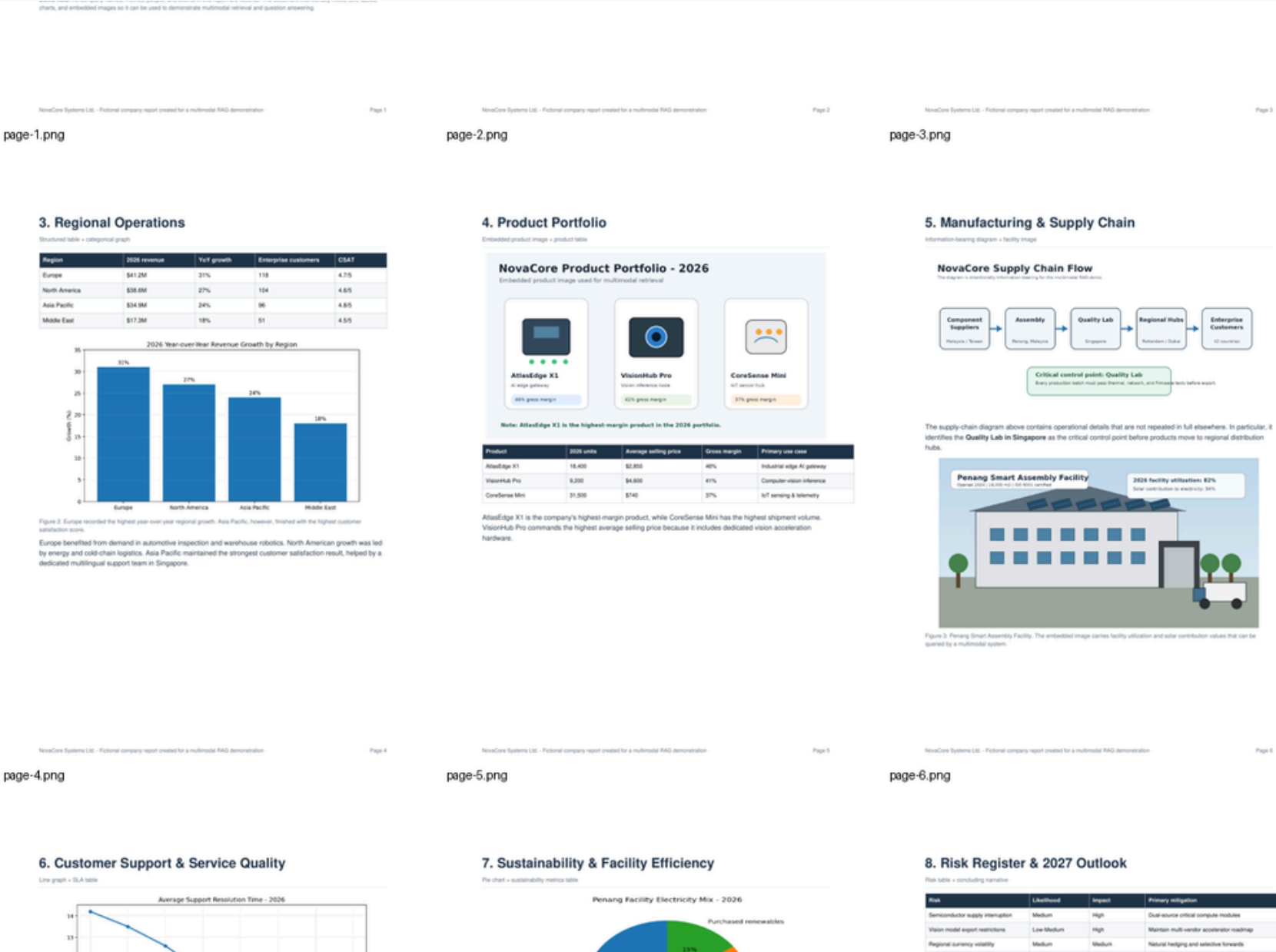
**“Convert every useful modality into something retrievable, then give the final model the original evidence it needs to answer accurately.”**

# Running example: NovaCore FY2026 report

A fictional company report designed specifically for a multimodal RAG demo.

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DEMO DOCUMENT



## Company snapshot

HQ

Singapore

2026 Revenue

\$132.0M

Employees

1,480

Operating Margin

18.4%

Sector

Industrial AI & IoT

Markets

42 countries

FICTIONAL DATA



# Example 1 — Text retrieval

The easiest case: facts already exist in normal narrative text.

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MODALITY EXAMPLE — TEXT

## Retrieved paragraph

*“FY2026 revenue reached USD 132.0 million, up from USD 97.5 million in FY2025. Europe became the fastest-growing region...”*

PAGE 2



QUESTION

What was NovaCore’s FY2026 revenue?

ANSWER

\$132.0 million (Page 2).

# Example 2 — Table retrieval

Keep rows and columns together so relationships remain meaningful.

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MODALITY EXAMPLE — TABLE

| Region        | 2026 revenue | YoY growth | Customers | CSAT  |
|---------------|--------------|------------|-----------|-------|
| Europe        | \$41.2M      | 31%        | 118       | 4.7/5 |
| North America | \$38.6M      | 27%        | 104       | 4.6/5 |
| Asia Pacific  | \$34.9M      | 24%        | 96        | 4.8/5 |
| Middle East   | \$17.3M      | 18%        | 51        | 4.5/5 |



**Question A**  
Which region grew fastest?



**Question B**  
Which region had the highest CSAT?

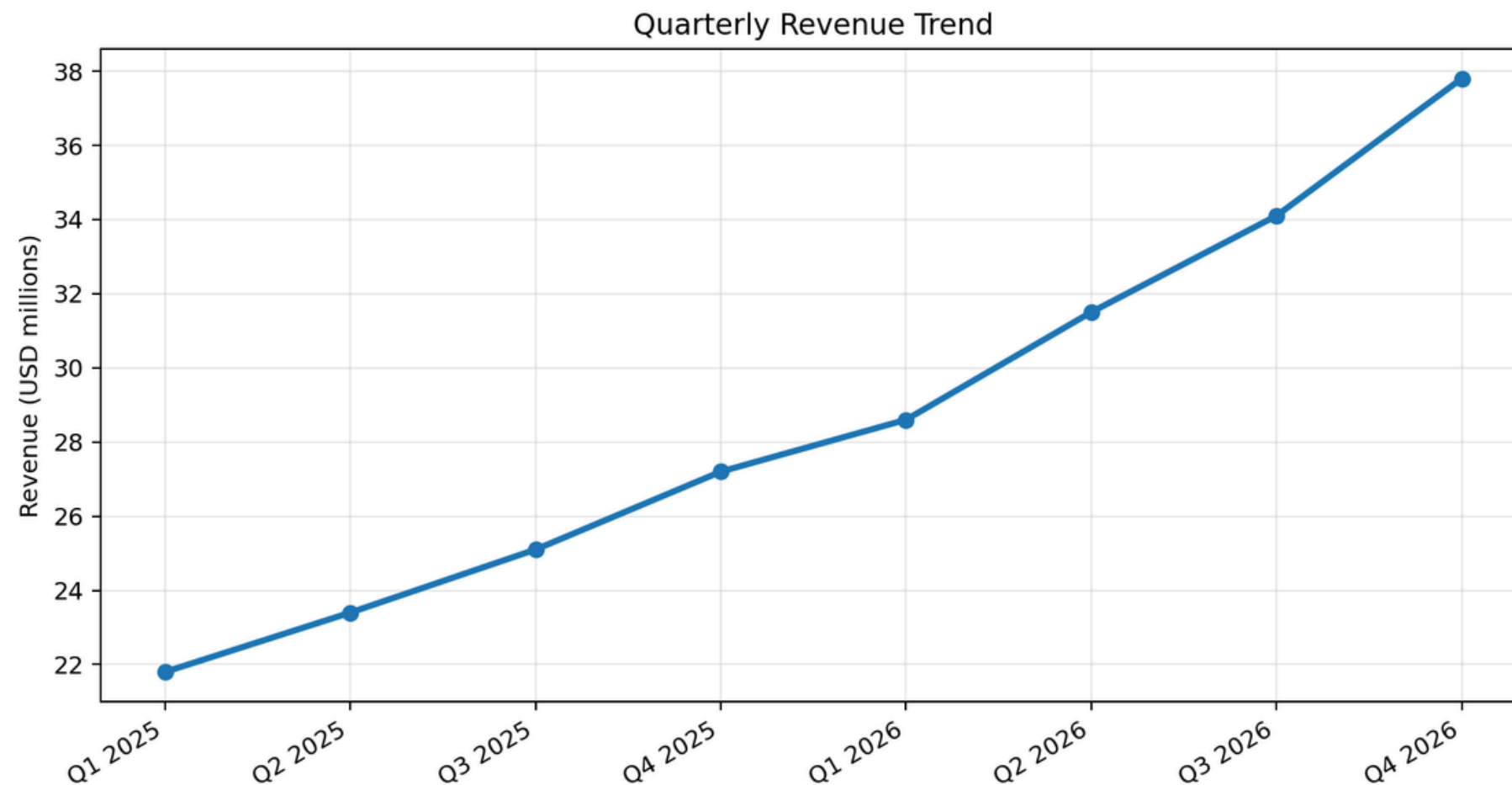
QUESTION Which region grew fastest?

ANSWER Europe — 31% YoY. (Asia Pacific had the highest CSAT: 4.8/5.)

# Example 3 — Graph / chart retrieval

A graph can contain the answer even when no sentence states it explicitly.

## 7 MODALITY EXAMPLE — GRAPH



V

### What the VLM summary might say

“Quarterly revenue increases throughout the period and peaks in Q4 2026 at about \$37.8M.”

P

### What we store

The summary is embedded in Pinecone, while the original chart image is retained for final visual reasoning.

QUESTION

Which quarter had the highest revenue?

ANSWER

Q4 2026 — approximately \$37.8M.



# Example 4 — Diagram retrieval

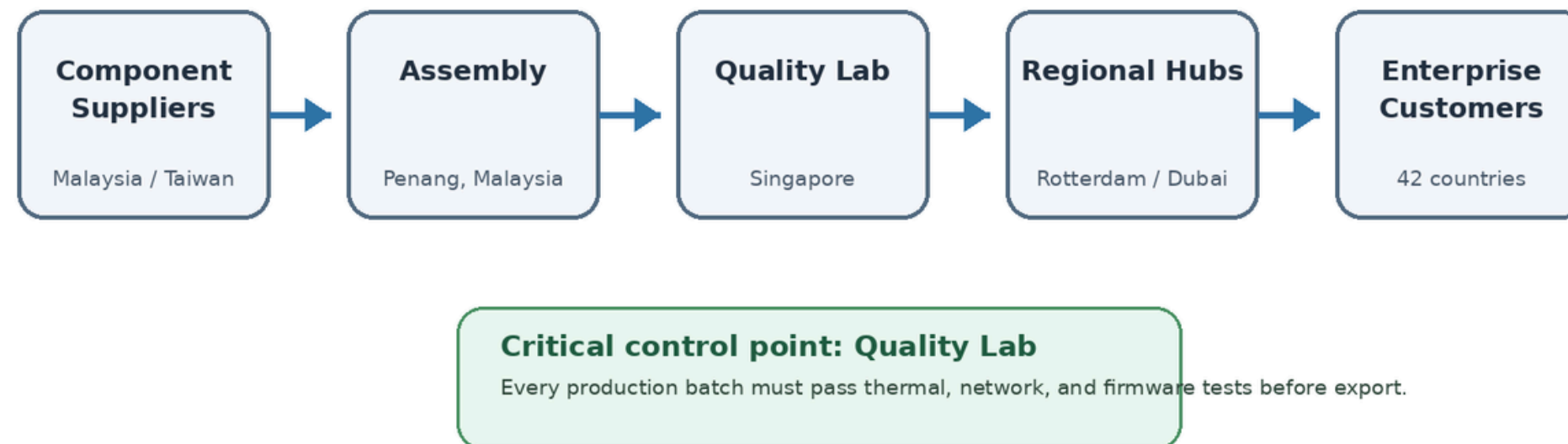
Relationships and flow often live inside diagrams—not in paragraph text.

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## MODALITY EXAMPLE — DIAGRAM

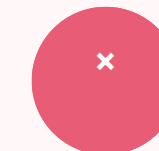
### NovaCore Supply Chain Flow

The diagram is intentionally information-bearing for the multimodal RAG demo.



### Visual fact

The diagram identifies the Quality Lab in Singapore as the critical quality-control point before regional distribution.



### Why text-only RAG can miss it

The meaning depends on boxes, arrows, order, and labels inside the image.

QUESTION

Where is the critical quality-control point?

ANSWER

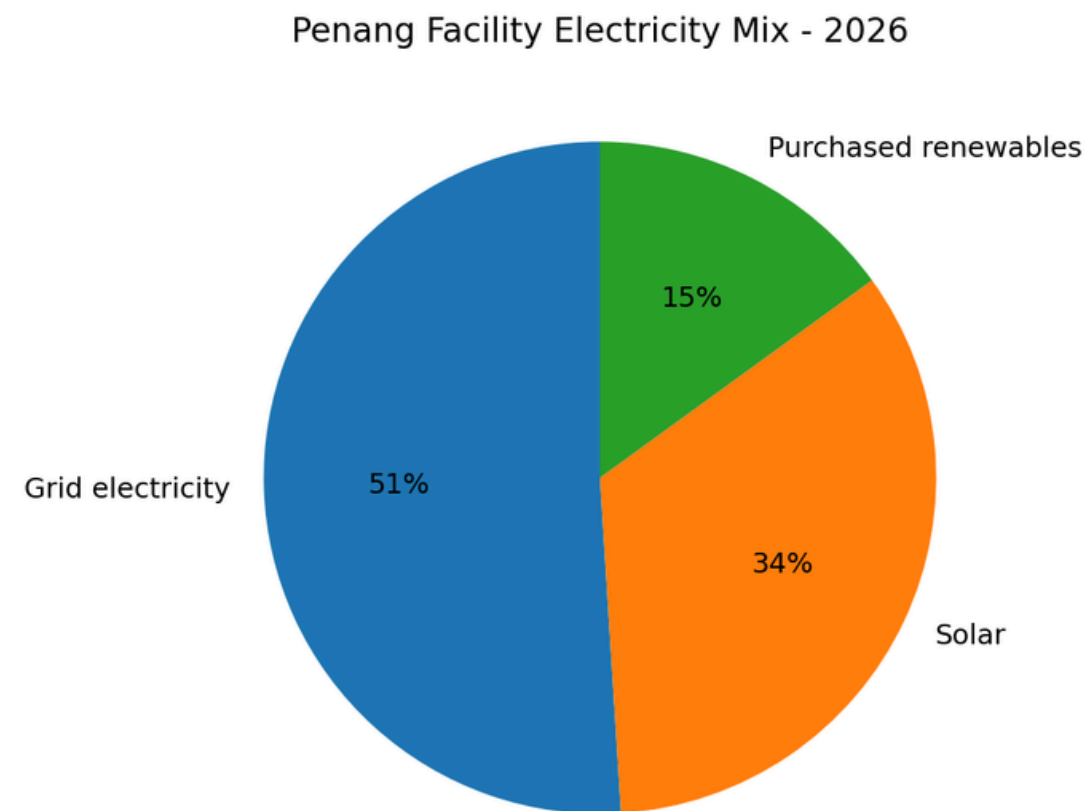
Quality Lab in Singapore.

# Example 5 — Pie chart retrieval

Visual proportions are a natural multimodal RAG question.

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## MODALITY EXAMPLE — CHART



### What can be answered from this visual?

%

- Solar share
- Purchased renewable share
- Grid electricity share
- Largest electricity source

The chart itself is the primary evidence.

QUESTION

What percentage of Penang electricity came from solar?

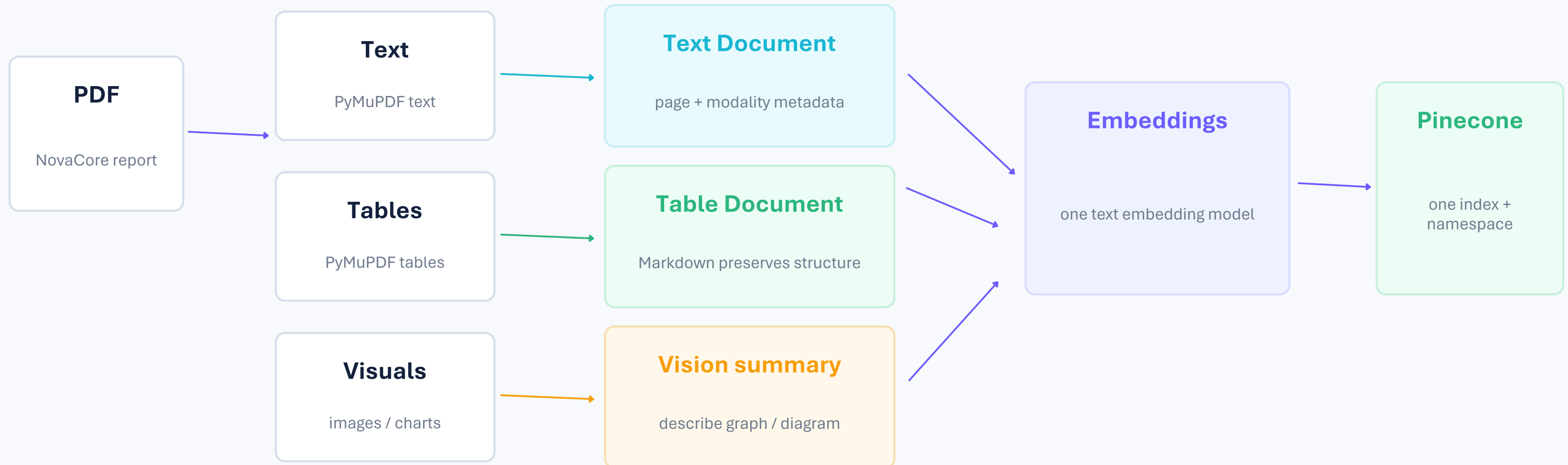
ANSWER

34%.

# Step 1 — Multimodal ingestion pipeline

Turn one PDF into modality-aware LangChain Documents.

## 10 INGESTION



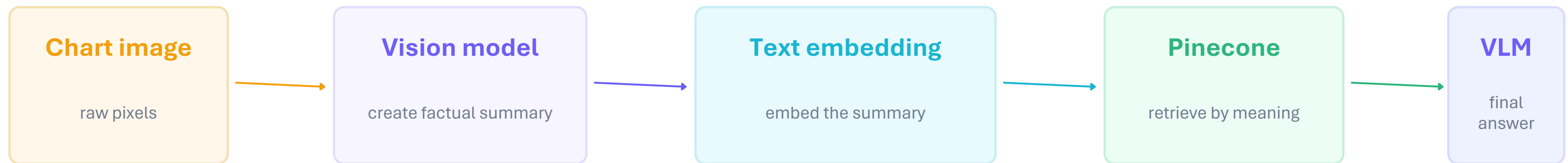
**Important: the raw visual is NOT discarded. We keep its path/reference for final answer generation.**

# Why summarize visuals before embedding?

A simple strategy: search with text vectors, answer with original visuals.

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## KEY DESIGN PATTERN



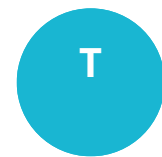
Store together

```
page_content = "Revenue rises every quarter and peaks in Q4 2026..."  
  
metadata = { page: 3, modality: "visual", image_path: ".../revenue_trend.png" }
```

# What goes into Pinecone?

One index can store searchable representations of every modality.

## 12 VECTOR STORE



### Text vector

page\_content:  
“FY2026 revenue reached USD  
132.0M...”

metadata:  
page: 2  
modality: text  
source: NovaCore PDF



### Table vector

page\_content:  
“| Europe | \$41.2M | 31% | ... |”

metadata:  
page: 4  
modality: table  
table\_number: 1



### Visual vector

page\_content:  
“Support time falls from 14.2h to  
8.1h...”

metadata:  
page: 7  
modality: visual  
image\_path: ...

All three are embedded with the same text embedding model → stored under one Pinecone namespace.



## Step 2 — Retrieval in action

The query retrieves a visual summary first, then we load the original graph.

### 13 QUERY-TIME FLOW

“How did support resolution time change from January to August?”

Pinecone Retriever

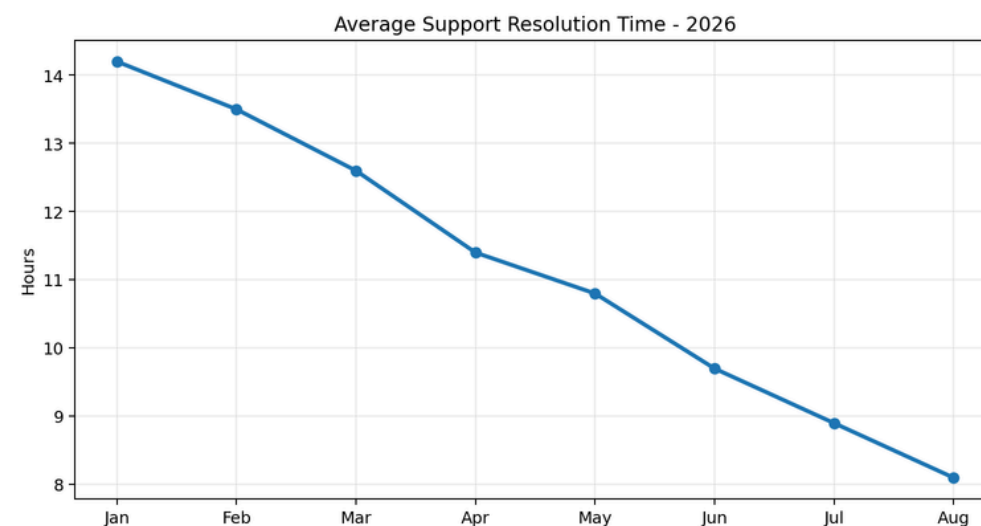
top-k semantic search

Visual summary

“14.2h → 8.1h”

Load image

metadata path



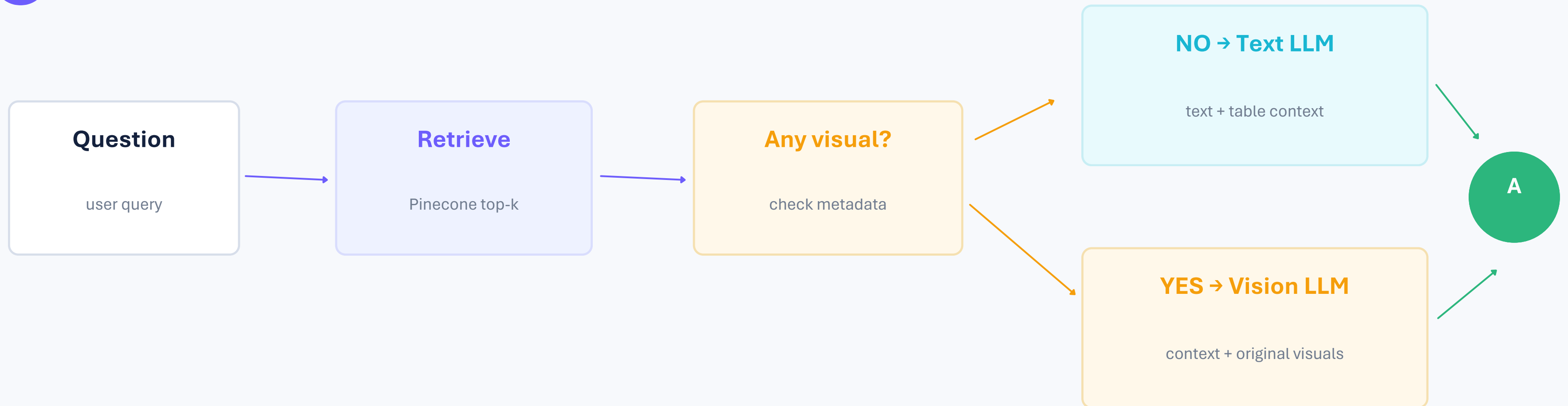
### Final grounded answer

Average resolution time fell from 14.2 hours in January to 8.1 hours in August — a reduction of about 43%.

# Step 3 — Keep query-time routing simple

No LangGraph needed for the first implementation.

## 15 GENERATION



# Final end-to-end architecture

The complete simple multimodal RAG pipeline you can implement in the notebook.

## 18 END-TO-END

